INCH-POUND
MIL-I-24768/7
25 February 1992
SUPERSEDING
MIL-P-19161A
12 December 1956

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, GLASS-CLOTH, POLYTETRAFLUOROETHYLENE-RESIN (GTE)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of a glass-cloth base or filler properly impregnated and bonded with polytetrafluoroethylene-resin compound or binder and processed to conform to this specification.

Intended use: The laminated material covered by this specification is intended for use in electronic applications requiring very low dielectric losses and very high volume resistivity and surface resistance.

Forms: Sheets are the only recommended form of laminated material covered by this specification.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

Length and width: Manufacturer's standard sizes between 16 and 50 inches in width and 36 and 96 inches in length will be acceptable.

Performance requirements: Performance requirements for laminated sheets shall be as specified in table II.

AMSC N/A FSC 5970
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TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inch)	Permissible variations (± inch)
M24768/7-S-1	0.031	0.0065
M24768/7-S-2	0.047	0.0075
M24768/7-S-3	0.062	0.0075
M24768/7-S-4	0.094	0.009
M24768/7-S-5	0.125	0.012
M24768/7-S-6	0.156	0.015
M24768/7-S-7	0.188	0.019
M24768/7-S-8	0.219	0.021
M24768/7-S-9	0.250	0.022
M24768/7-S-10	–[1]	–[2]

[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.025 inch thick sheet is: M24768/7-S-10-0.025.

[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

The following performance requirements do not apply to this plastic laminate:

a. Specific gravity
b. Dielectric strength
c. Bonding strength
d. Flammability
e. Compressive strength
f. Silicone content
g. Tracking resistance
h. Thermal endurance.

Testing: The laminated material covered by this specification shall be tested under qualification inspection requirements.

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TABLE II. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]					
				0.031	0.062	0.094	0.125	0.188	0.250
Dielectric breakdown voltage	A D-48/50	4.6.4	Min. kV	32.0 20.0	32.0 20.0	32.0 20.0	32.0 20.0	32.0 20.0	32.0 20.0
Dielectric constant	D-24/23	4.6.6	Max.	2.8	2.8	2.8	2.8		
Dissipation factor	D-24/23	4.6.6	Max.	0.005	0.005	0.005	0.005		
Volume resistivity	C-96/35/90	4.6.7	Min. megohm cm				10^7		
Surface resistance	C-96/35/90	4.6.8	Min. megohm				10,000		
Arc resistance	D-48/50	4.6.9	Min. seconds		180	180	180		
Tracking resistance	A	4.6.10	Min. minutes						
Impact strength Lengthwise: Crosswise:	E-48/50	4.6.11	Min. ft–lbs. per inch				10.0 6.0	10.0 6.0	10.0 6.0
Flexural strength Lengthwise: Crosswise:	A	4.6.12	Min. psi	15,000 10,000	15,000 10.000	15,000 10,000	15,000 10,000	15,000 10,000	15,000 10,000

See footnotes at the end of table.

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TABLE II. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]					
				0.031	0.062	0.094	0.125	0.188	0.250
Thermal endurance Flexural strength:		4.6.13	Min. psi						
Bonding strength	A D-48/50	4.6.15	Min. pounds						
Water absorption	D_1-24/23	4.6.17	Max. percent	0.20	0.10	0.08	0.08	0.08	0.08
Silicone content	E-168/185	4.6.18	Max. ppm						
Flame resistance		4.6.19							
Ignition time:	A		Min. sec.						
Burning time:			Max. sec.						

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.

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Custodians:
Army – ER
Navy – SH
Air Force – 20

Review activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User activities:
Army – ME
Navy – MC, AS

Preparing activity:
Navy – SH
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